A sufficiently saturated ordered system realises the criterion for every element

Silvia Barbina, Enrique Casanovas

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Editorial note. The counted statement is the author's environment at source0.tex lines 894-895 and its proof environment opens at line 896; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

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Editorial scope. One original result of the pinned author manuscript, counted once: in a sufficiently saturated ordered system the closure of a generated subsystem is itself (author label June_25_2024_4).

Allowed standard prerequisites (editorial).

- Saturation of the ordered structure. The system is kappa-saturated as a structure in the language with the ordering, so types over small sets are realised.. It produces the element above the subsystem.
- Binary tree above an element. In an omega-saturated ordered system there is a binary tree of height omega above any element.. It supplies the tree that the criterion of the previous proposition needs.
- Generated subsystem. A = <A> means A is generated by itself.. It is the hypothesis on the subsystem.

Corollary 0.1 Let $<$ be an HF-ordering of the infinite STS M and assume M is ω-saturated as an $(R, <)$-structure. For every $a \in M$ there is a binary tree $T = \{a_s \mid s \in 2^{<\omega}\}$ in M such that

1. $a_s > a$ for every s
2. $a_s > a_t$ if $s \subsetneq t$.

Proposition 0.2 Assume M is an infinite unconfined STS, $A = \langle A \rangle \subseteq M$ and $a \in M \setminus A$. The following are equivalent:

1. $\mathrm{cl}_<(a) = \{a\}$ for some HF-ordering of M with initial segment A
2. $A \leq Aa \leq M$
3. There are $M' \succeq M$, an HF-ordering $<'$ of M' and a binary tree $T = \{a_s \mid s \in 2^{<\omega}\}$ in M' such that $\mathrm{cl}_{<'}(A) = A$, $a = a_\emptyset$, $T >' A$ and $a_s >' a_{s0}, a_{s1}$ for every $s \in 2^{<\omega}$.

Moreover, these conditions imply $a \leq M$.

Corollary 0.3 Let M be an infinite unconfined STS and $A \subseteq M$. If $<$ is an HF-ordering of M such that $(M, <)$ is κ-saturated for some $\kappa > |A|$ and $\mathrm{cl}_<(A) = A$, then there is some $a \in M \setminus A$ such that $Aa \leq M$.

Proof. By κ-saturation there is some $b \in M$ such that $b > A$. By Corollary 0.1 we find a binary tree $T > b$ of height ω in M and by the proof of $3 \Rightarrow 2$ in Proposition 0.2, the root a of the tree verifies $Aa \leq M$. □